

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS 2026 3 QUESTIONS 5% WEIGHTAGE

↑50.15%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Variable Separable & Homogeneous DE	26 PYQs	Separable DE
Moderate	Linear Differential Equations & Integrating Factor	5 PYQs	Linear DE

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Variable Separable & Homogeneous DE	$\int \frac{dy}{g(y)} = \int f(x) dx + C$, $\quad \text{For homogeneous: substitute } y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$
Linear Differential Equations & Integrating Factor	$\frac{dy}{dx} + P(x)y = Q(x) \implies \text{I.F.} = e^{\int P(x) dx}$, $\quad y \cdot \text{I.F.} = \int Q(x) \cdot \text{I.F.} dx + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Variable Separable & Homogeneous DE (Very High Yield — 26 PYQs)

Core Principle: Classification of order and degree, separation of variables, and reduction of homogeneous equations via $y = vx$.

■ **Caution / JEE Pitfall:** Degree of a differential equation is defined ONLY when the equation is a polynomial in its derivatives (no fractional powers or $\sin(dy/dx)$).

■ **Speed Trick:** Equations of the form $\frac{dy}{dx} = f(ax + by + c)$ can be solved by substituting $z = ax + by + c$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Sum of the squares of the order and degree (if defined) of a differential equation $2y'' + (y')^2 = \sqrt{(y'')^2 - 3}$ is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): " $\int xf(x)dx + (f(x)/2) = 0$, then $f(x)$ is equal to..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "Integrating factor of the differential equation $(1+x^2)(dy/dx) + xy = 1$ is..." [Ans: D]

3.2 Linear Differential Equations & Integrating Factor (Moderate Yield — 5 PYQs)

Core Principle: Standard first-order linear differential equations and integrating factor technique for $y(x)$ and $x(y)$.

■ **Caution / JEE Pitfall:** If the equation has x $\frac{dy}{dx}$, remember to divide by x throughout first so the coefficient of $\frac{dy}{dx}$ is 1!

■ **Speed Trick:** If the equation is non-linear in y , check if it is linear in x : $(\frac{dx}{dy}) + P(y)x = Q(y)$ implies I.F. = $e^{\int P(y)dy}$.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "Integrating factor of the differential equation $(1+x^2)(\frac{dy}{dx})+xy=1$ is..." [Ans: D]
- [2024 Shift] Q#3 (MCQ): "The family of curves whose x and y intercepts of a tangent at any point are respectively double the x and y coordinates of that point..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "The integrating factor of the differential equation $(2x+3y^2)dy=ydx (y>0)$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Sum of the squares of the order and degree (if defined) of a differential equation $2y''+(y'')^2=\sqrt{(y''')^3}$ is

- (A) 3
- (B) 20 [Correct]
- (C) 8
- (D) 16

Correct Answer: B

Solution Analysis: The given differential equation is $2y''+(y'')^2=\sqrt{(y''')^3}$. The highest order derivative present is y''' , so the order of the differential equation is 2. To find the degree, the equation must be written as a polynomial in derivatives. Since y''' is inside a square root, we first remove the radical by squaring both sides: $(2y''+(y'')^2)^2=y'''^3$. Expanding, $4(y'')^2+4y''(y'')^2+(y'')^4=y'''^3$. Bring all terms to one side: $(y''')^3-4(y'')^2-4y''(y'')^2-(y'')^4=0$. Now it is a polynomial in derivatives. The highest order deriva

Benchmark Problem 2: KCET 2024 (Q#3)

The family of curves whose x and y intercepts of a tangent at any point are respectively double the x and y coordinates of that point is

- (A) $xy=C$
- (B) $x^2+y^2=C$
- (C) $x^2-y^2=C$
- (D) $(y/x)=C$

Correct Answer: None

Solution Analysis: Let the point on the curve be (x, y) . According to the question, the equation of the tangent will be $\Rightarrow \frac{x}{2x} + \frac{y}{2y} = 1 \Rightarrow \frac{1}{2} + \frac{1}{2} = 1$. So, the slope of tangent will be $-\frac{y}{x}$. Now, $\frac{dy}{dx} = -\frac{y}{x}$. On integrating both sides, we get $\int \frac{dy}{y} = -\int \frac{dx}{x} \Rightarrow \ln y = -\ln x + \ln C \Rightarrow \ln y = \ln \frac{C}{x} \Rightarrow y = \frac{C}{x}$.